

Macroeconomics Practice Questions

Solutions in LaTeX

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1 IMAGE 1: Basic Macroeconomics Concepts

1.1 Question 1: Consumption Function and Determinants

The Consumption Function

The consumption function describes the relationship between consumption expenditure and disposable income:

$$C = a + bY_d \quad (1)$$

Where:

- C = Total consumption expenditure
- a = Autonomous consumption (consumption when income is zero)
- b = Marginal Propensity to Consume (MPC)
- Y_d = Disposable income

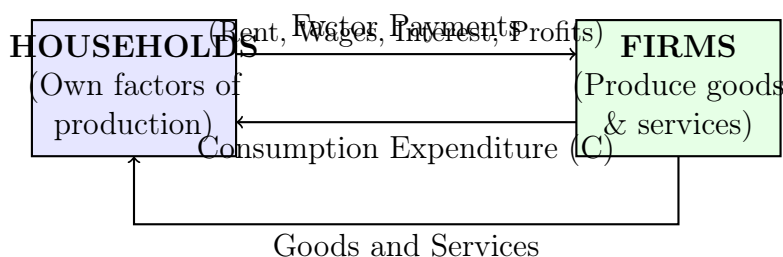
Determinants of Consumption Expenditure

Factor	Explanation
Disposable Income	Primary determinant; as income rises, consumption rises (but less than proportionately)
Wealth	Higher wealth increases consumption (wealth effect)
Interest Rates	Higher rates encourage saving, reducing current consumption
Expectations	Optimistic future expectations increase current spending
Inflation	Expected inflation may increase current consumption
Distribution of Income	More equal distribution increases aggregate consumption
Demographic Factors	Age structure, family size affect consumption patterns
Government Policies	Taxation and social security affect disposable income

1.2 Question 2: Circular Flow of Income in a Closed Economy

The Circular Flow Model

The circular flow model illustrates how money, goods, and services flow between households and firms in a closed economy (no foreign trade).



Leakages (Withdrawals)

These reduce the flow of income:

- **Saving (S)** – Income not spent on consumption
- **Taxes (T)** – In a three-sector model with government
- **Imports (M)** – In an open economy

Injections

These add to the flow of income:

- **Investment (I)** – Spending on capital goods
- **Government Spending (G)** – In a three-sector model
- **Exports (X)** – In an open economy

Equilibrium Condition

$$\text{Leakages} = \text{Injections} \quad \Rightarrow \quad S + T + M = I + G + X \quad (2)$$

1.3 Question 3: Nominal vs. Real National Income

Aspect	Nominal National Income	Real National Income
Definition	National income measured at current market prices	National income measured at constant (base year) prices
Formula	Current quantity $\times$ Current prices	Current quantity $\times$ Base year prices
Adjustment	No adjustment for inflation	Adjusted for inflation using price index
Comparison	Cannot compare across time periods	Enables meaningful comparison over time
Value	May rise due to price increases alone	Reflects actual change in output/volume

Relationship

$$\text{Real National Income} = \frac{\text{Nominal National Income}}{\text{Price Index}} \times 100 \quad (3)$$

Uses of National Income Statistics

1. **Economic Planning** – Helps government formulate development plans
2. **Standard of Living Comparison** – Per capita income indicates living standards
3. **International Comparisons** – Compare economic performance across countries
4. **Policy Formulation** – Guides fiscal and monetary policies
5. **Resource Allocation** – Shows sectoral contributions to GDP
6. **Investment Decisions** – Helps investors assess market potential
7. **Employment Analysis** – Relationship with employment levels
8. **Income Distribution** – Analysis of inequality

1.4 Question 4: Expenditure Approach and Standard of Living**a) Expenditure Approach to National Income Measurement**

The expenditure approach calculates GDP by summing all final expenditures:

$$GDP = C + I + G + (X - M) \quad (4)$$

Component Description		
C	(Consumption)	Household spending on goods and services (durable, non-durable, services)
I	(Investment)	Business spending on capital goods, residential construction, inventory changes
G	(Government)	Government spending on goods and services (excluding transfers)
X	(Exports)	Goods and services sold to foreign countries
M	(Imports)	Goods and services purchased from foreign countries (subtracted)

b) Relationship Between National Income and Standard of Living

Positive Aspects	Limitations
Higher per capita income enables better access to goods/services	Does not account for income distribution (inequality)
Allows investment in healthcare, education, infrastructure	Ignores non-market activities (household work, volunteer services)
Reduces poverty levels	Environmental costs not deducted (pollution, resource depletion)
Provides employment opportunities	Does not measure leisure time or working conditions
Enables technological advancement	Quality improvements in goods not fully captured
Improves housing and nutrition	Underground economy excluded

Conclusion: National income is a useful but imperfect indicator of standard of living. The Human Development Index (HDI) provides a more comprehensive measure.

2 IMAGE 2: Consumption Theories and Calculations

2.1 Question 1: Relative Income Hypothesis (Duesenberry)

Developed by: James Duesenberry (1949)

Core Propositions

1. The Ratchet Effect:

- Consumers resist reducing their consumption levels when income falls
- Once a higher standard of living is achieved, people maintain it even if income declines
- Consumption depends on **past peak income** rather than current income
- Explains why consumption is more stable than income during recessions

2. The Demonstration Effect (Keeping up with the Joneses):

- An individual's consumption is influenced by the consumption patterns of others in their social group
- People try to maintain their relative position in the income distribution
- Consumption depends on **relative income** position, not absolute income

Consumption Function

$$\frac{C_t}{Y_t} = \alpha + \beta \left(\frac{Y_0}{Y_t} \right) \quad (5)$$

Where Y_0 = previous peak income

Critical Evaluation

Strengths	Weaknesses
Explains short-run rigidity of consumption	Difficult to empirically verify
Accounts for consumer behavior during business cycles	Ignores role of liquid assets
Recognizes social/psychological factors in consumption	Assumes consumption is irreversible
Better explains reality than absolute income hypothesis	Does not explain saving behavior adequately
Explains why APC may rise or fall	Limited predictive power

2.2 Question 2: Three-Sector Economy Calculations

Given:

- Autonomous consumption (a) = #95 million
- Investment (I) = #520 million
- Government expenditure (G) = #250 million
- MPC (b) = 0.8

The Consumption Function: $C = 95 + 0.8Y$

Equilibrium Condition: $Y = C + I + G$

a) Equilibrium Level of National Income

$$Y = C + I + G \quad (6)$$

$$Y = (95 + 0.8Y) + 520 + 250 \quad (7)$$

$$Y = 865 + 0.8Y \quad (8)$$

$$Y - 0.8Y = 865 \quad (9)$$

$$0.2Y = 865 \quad (10)$$

$$Y = \#4,325 \text{ million} \quad (11)$$

b) Equilibrium Consumption Expenditure

$$C = 95 + 0.8Y \quad (12)$$

$$C = 95 + 0.8(4,325) \quad (13)$$

$$C = 95 + 3,460 \quad (14)$$

$$C = \#3,555 \text{ million} \quad (15)$$

Verification: $C + I + G = 3,555 + 520 + 250 = 4,325 = Y \checkmark$

c) Change in National Income when Investment Increases by 20%

Step 1: Calculate new investment

$$\Delta I = 20\% \times 520 = 0.20 \times 520 = \#104 \text{ million} \quad (16)$$

$$I_{new} = 520 + 104 = \#624 \text{ million} \quad (17)$$

Step 2: Calculate the Multiplier

$$k = \frac{1}{1 - MPC} = \frac{1}{1 - 0.8} = \frac{1}{0.2} = 5 \quad (18)$$

Step 3: Calculate change in national income

$$\Delta Y = k \times \Delta I = 5 \times 104 = \#520 \text{ million} \quad (19)$$

New Equilibrium Income: $4,325 + 520 = \#4,845 \text{ million}$

3 IMAGE 3 & 4: Money, Wages and Banking

3.1 Question 1: Money Wages vs. Real Wages

Money Wages	(Nominal Wages)	Real Wages
Wages measured in terms	measured in monetary terms	Wages measured in terms of purchasing power
The actual amount of currency received		The quantity of goods and services wages can buy
Not adjusted for inflation		Adjusted for inflation using price index
Example: #50,000 per month		Example: #50,000 can buy 2 bags of rice

Relationship

$$\text{Real Wage} = \frac{\text{Money Wage}}{\text{Price Level}} \times 100 \quad (20)$$

Effects of Rising Urban Wages on Nigerian Economy

Positive Effects	Negative Effects
Increased Purchasing Power – Higher living standards for urban workers	Rural-Urban Migration – Brain drain from agricultural sector
Increased Consumption – Boosts aggregate demand and economic growth	Inflationary Pressure – Wage-price spiral if not matched by productivity
Attracts Investment – Skilled labor attracts foreign direct investment	Income Inequality – Widening gap between urban and rural areas
Improved Productivity – Better-paid workers are more motivated	Cost-Push Inflation – Higher production costs reduce competitiveness
Tax Revenue – Higher income tax revenue for government	Unemployment – If wages rise above market-clearing level
Human Capital Development – Enables education and skill acquisition	Dutch Disease – Over-reliance on urban sectors, neglect of agriculture

3.2 Question 2: Motives for Holding Money (Keynesian Liquidity Preference)

1. Transactions Motive:

- Money held for daily transactions and regular payments
- Directly related to income level
- $L_t = kY$ (where k is a constant)

2. Precautionary Motive:

- Money held for unforeseen emergencies and unexpected expenses
- Depends on income level and uncertainty
- $L_p = f(Y, \text{uncertainty})$

3. Speculative Motive:

- Money held to take advantage of future changes in interest rates/bond prices
- Inversely related to interest rates
- $L_s = f(r)$ where $\frac{dL_s}{dr} < 0$

Total Demand for Money

$$L = L_t + L_p + L_s = L_1(Y) + L_2(r) \quad (21)$$

3.3 Question 3: Central Bank of Nigeria (CBN) and Economic Development

Function	Contribution to Development
Monetary Policy	Controls money supply and interest rates to maintain price stability, creating conducive environment for investment
Banker to Government	Manages government accounts, facilitates deficit financing for development projects
Banker to Banks	Acts as lender of last resort, ensures banking system stability
Foreign Exchange Management	Manages external reserves, stabilizes exchange rate, facilitates international trade
Development Finance	Through institutions like Bank of Industry (BOI), NEXIM Bank, provides long-term credit to key sectors
Financial Inclusion	Promotes access to financial services, reducing poverty
Payment System Oversight	Modernizes payment infrastructure (eNaira), reduces transaction costs
Prudential Regulation	Supervises banks to prevent failures and protect depositors

Specific CBN Interventions:

- Anchor Borrowers' Programme (ABP) for agriculture
- Real Sector Support Facility (RSSF)
- COVID-19 intervention funds
- Micro, Small and Medium Enterprises Development Fund (MSMEDF)

3.4 Question 4: Problems of Trade by Barter

Barter System: Direct exchange of goods for goods without using money

Problem	Explanation
Double Coincidence of Wants	Both parties must want what the other has; difficult to find matching needs
Indivisibility	Some goods cannot be divided into small units (e.g., a cow cannot be split for small transactions)
Lack of Common Measure of Value	No standard unit to compare values of different goods
Difficulty in Storing Wealth	Perishable goods lose value; no efficient way to save purchasing power
No Standard for Deferred Payments	Impossible to lend or borrow, as future repayment value is uncertain
Transportation Costs	Bulky goods are expensive and difficult to transport for exchange
No Specialization	Everyone must produce a variety of goods to trade, reducing efficiency
Transaction Costs	High time and effort spent searching for trading partners

3.5 Question 5: Quantity Theory of Money**a) Irving Fisher's Equation of Exchange**

Original Form:

$$MV = PT \quad (22)$$

Where:

Symbol	Meaning
M	Money supply in circulation
V	Velocity of circulation (average times money changes hands)
P	General price level
T	Total volume of transactions

Fisher's Modification (Income Version):

$$MV = PY \tag{23}$$

Where Y = Real national income (output), replacing T

Assumptions:

- V is constant (determined by institutional factors)
- Y is constant at full employment
- Therefore, changes in M directly cause proportional changes in P

b) Critique of the Quantity Theory of Money

Criticism	Explanation
Velocity is not constant	V changes with financial innovations, payment habits, and interest rates
Full employment assumption unrealistic	Economies often operate below full employment; increased M can raise output
Direction of causation	May be reverse: price changes cause money supply changes, not vice versa
Hoarding	People may hold money idle, breaking the direct M - P relationship
Short-run vs. Long-run	Theory works better in long run; short-run price rigidities exist
Ignores other factors	Ignores cost-push inflation, structural factors, and expectations
Keynesian critique	In liquidity trap, increasing M has no effect on prices or output
Endogenous money	Modern banking creates money through lending, making M less controllable